COURSE 5 · ASTRONOMY

Astronomy

Visual companion and Symbols Sheets

One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

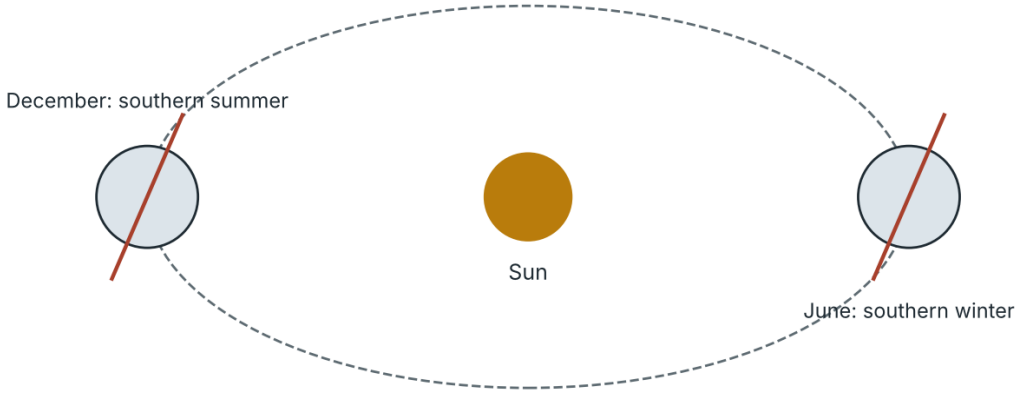
Keep it open while you listen, and beside you for the workbook.

Module 5.1 · Symbols and diagrams

The sky and the history of astronomy

5.1-1	Earth's axial tilt	$\varepsilon \approx 23.4^\circ$
5.1-2	Noon Sun altitude at the solstices	$h = 90^\circ - \lvert\text{latitude}\rvert \pm 23.4^\circ$
5.1-3	Synodic month	$P_{\text{phases}} \approx 29.5 \text{ days}$

The axis keeps pointing the same way in space (red, 23.4°). Which hemisphere leans towards the Sun changes through the year.

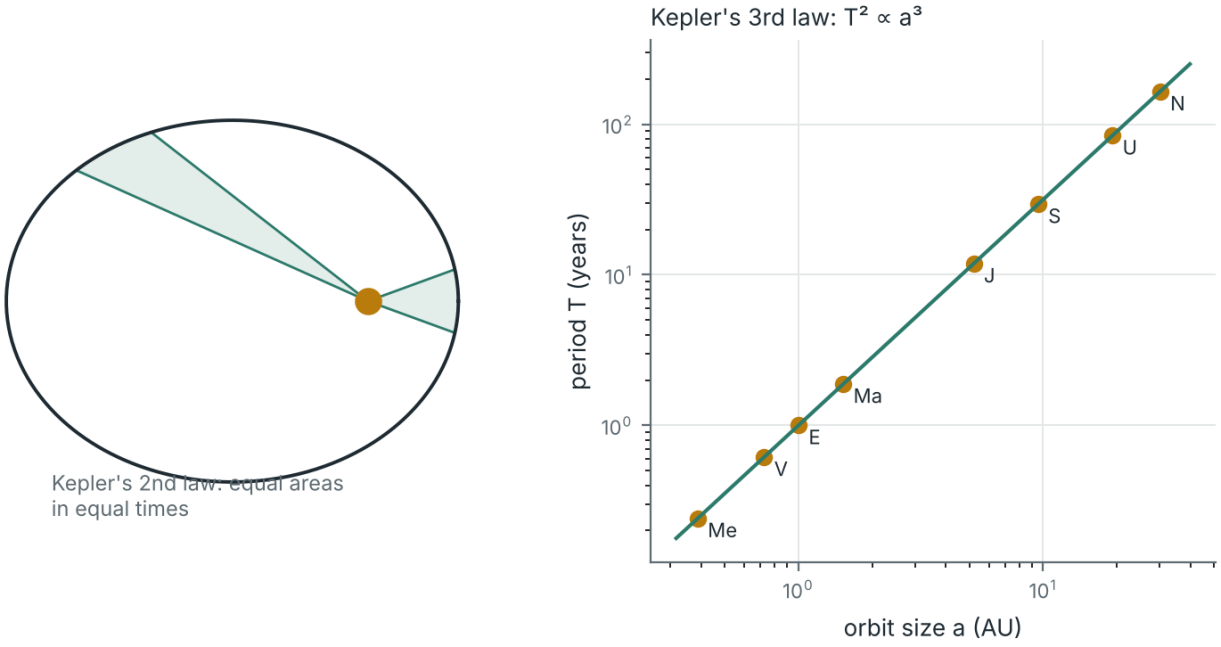


Seasons come from the tilt of Earth's axis, which points the same way in space all year.

Module 5.2 · Symbols and diagrams

Gravity and orbits

5.2-1	Kepler's third law (Solar System units)	$T^2 = a^3 \quad (T \text{ in years},\ a \text{ in AU})$
5.2-2	Universal gravitation	$F = \frac{GMm}{r^2}, \quad G = 6.67 \times 10^{-11}\ \text{N}\,\text{m}^2\,\text{kg}^{-2}$
5.2-3	Surface gravity	$g = \frac{GM}{R^2}$
5.2-4	Circular orbital speed	$v = \sqrt{\frac{GM}{r}}$
5.2-5	Kepler's third law (Newton's form)	$T^2 = \frac{4\pi^2}{GM} r^3$
5.2-6	Escape velocity	$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$

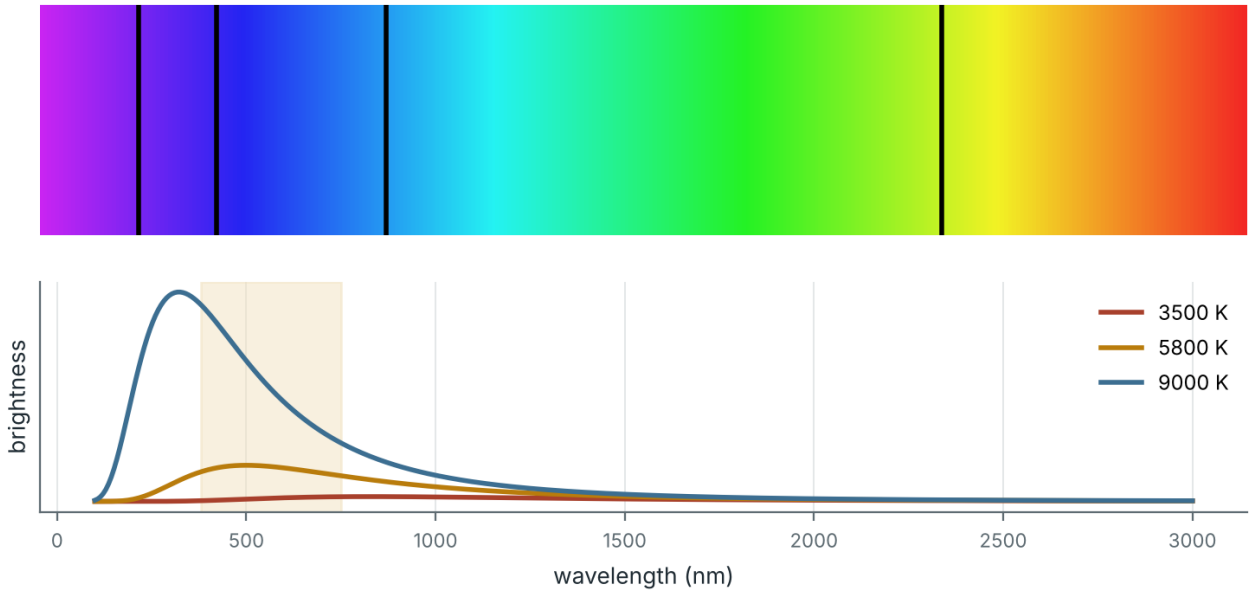


Left: equal areas in equal times. Right: the eight planets on a log–log plot follow $T^2 \propto a^3$.

Module 5.3 · Symbols and diagrams

Light as messenger

5.3-1	Inverse-square law	$b = \frac{L}{4\pi d^2}$
5.3-2	Parallax distance	$d\,(\mathrm{pc}) = \frac{1}{p\,(\mathrm{arcsec})}, \quad 1\ \mathrm{pc} \approx 3.26\ \mathrm{ly}$
5.3-3	Wien's law	$\lambda_{\max} = \frac{2.90 \times 10^{-3}\ \mathrm{m\,K}}{T}$
5.3-4	Doppler shift (v much less than c)	$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$

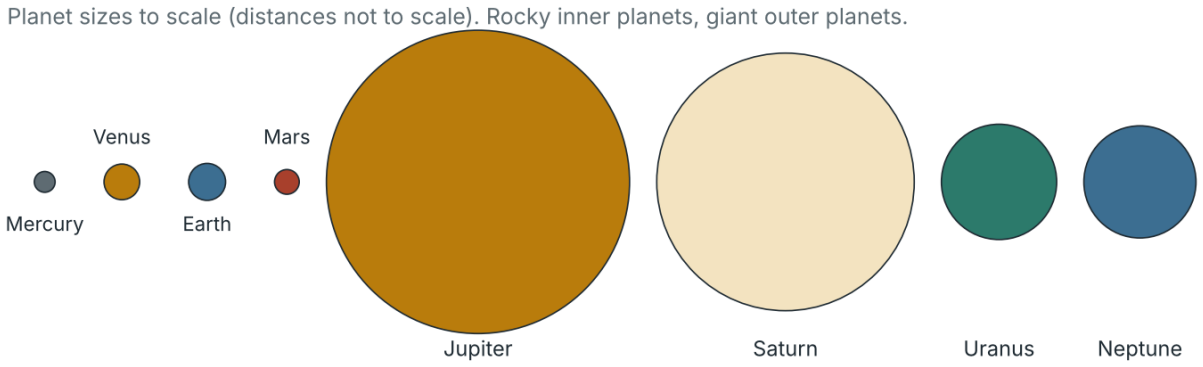


Top: absorption lines of hydrogen in a stellar spectrum. Bottom: hotter bodies are brighter and peak at shorter wavelengths (visible band shaded).

Module 5.4 · Symbols and diagrams

The Solar System

5.4-1	Solar power from mass loss	$P = \frac{\Delta m}{\Delta t} c^2 \approx 3.8 \times 10^{26}\ \text{W}$
5.4-2	Astronomical unit	$1\ \text{AU} \approx 1.50 \times 10^{11}\ \text{m}$
5.4-3	Transit depth	$\frac{\Delta F}{F} \approx \left(\frac{R_{\text{planet}}}{R_{\text{star}}}\right)^2$



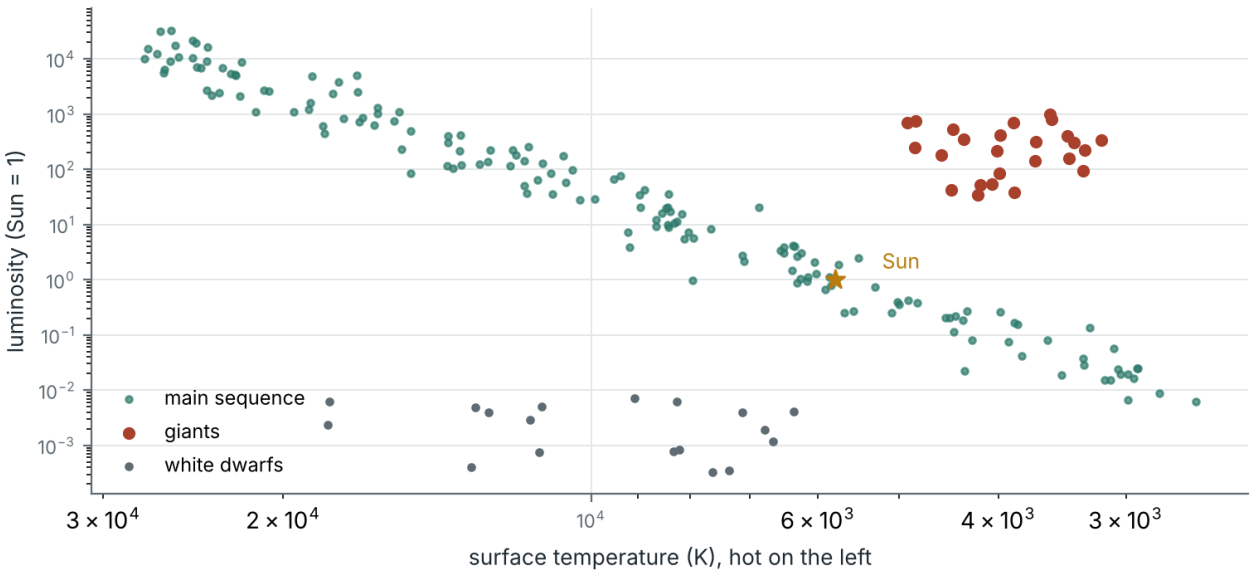
The planets' sizes to scale.

Planet	Distance (AU)	Type	Notable
Mercury	0.39	rocky	no atmosphere
Venus	0.72	rocky	runaway greenhouse
Earth	1.00	rocky	liquid water, life
Mars	1.52	rocky	ancient rivers
Jupiter	5.2	gas giant	largest planet
Saturn	9.5	gas giant	ring system
Uranus	19.2	ice giant	tilted 98°
Neptune	30.1	ice giant	fastest winds

Module 5.5 · Symbols and diagrams

Stars

5.5-1	Stefan–Boltzmann law	$L = 4\pi R^2 \sigma T^4, \quad \sigma = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}K^{-4}}$
5.5-2	Main-sequence lifetime	$t \approx 10^{10}\left(\frac{M}{M_\odot}\right)^{-2.5}\ \text{years}$
5.5-3	Hydrogen fusion (net)	$4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{He} + \text{energy}$

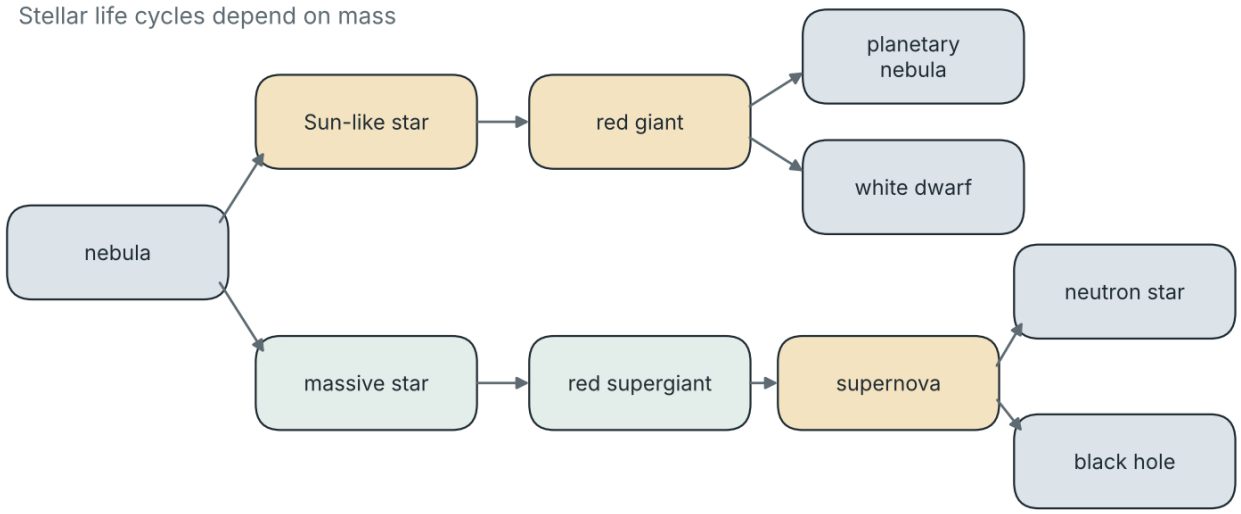


A schematic Hertzsprung–Russell diagram. Temperature increases to the left; luminosity is on a log scale.

Module 5.6 · Symbols and diagrams

Stellar death

5.6-1	Schwarzschild radius	$R_s = \frac{2GM}{c^2}$
5.6-2	Density	$\rho = \frac{M}{\frac{4}{3}\pi R^3}$
5.6-3	Chandrasekhar limit (white dwarf maximum)	$M_{\max} \approx 1.4\, M_\odot$

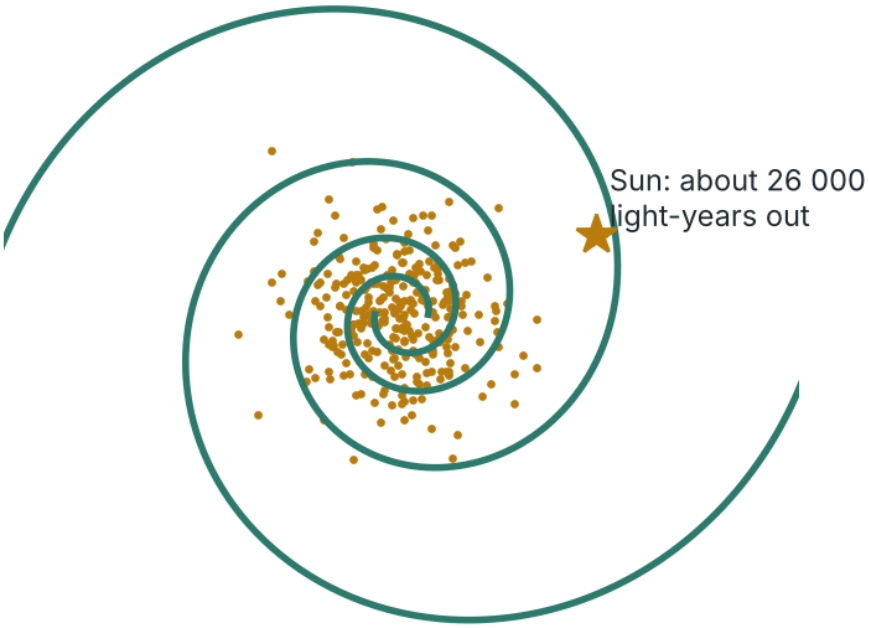


The life cycle of a star depends on its starting mass.

Module 5.7 · Symbols and diagrams

Galaxies

5.7-1	Orbital period around the galaxy	$T = \frac{2\pi r}{v}$
5.7-2	Mass inside an orbit	$M = \frac{v^2 r}{G}$
5.7-3	Light-year	$1\ \text{ly} \approx 9.46 \times 10^{15}\ \text{m}$



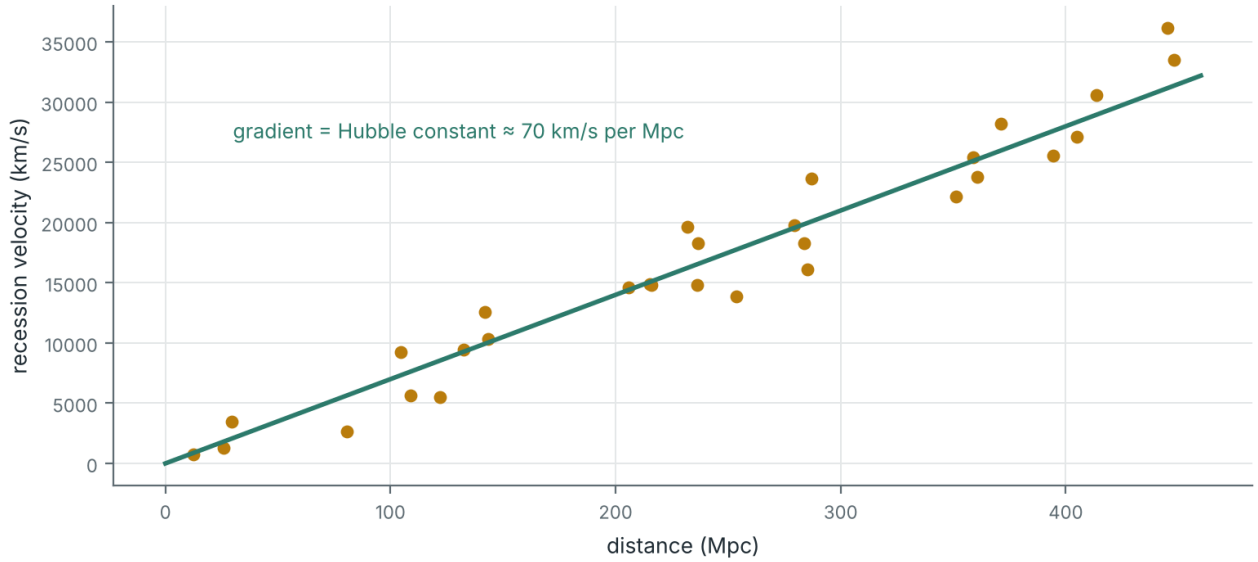
The Milky Way from above (schematic): a barred spiral about 100 000 light-years across

Schematic Milky Way seen from above, showing the Sun's position in a spiral arm.

Module 5.8 · Symbols and diagrams

Cosmology

5.8-1	Hubble's law	$v = H_0 d, \quad H_0 \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$
5.8-2	Hubble time	$t \approx \frac{1}{H_0}$
5.8-3	Composition today	$\Omega_b \approx 0.05, \quad \Omega_{dm} \approx 0.27, \quad \Omega_\Lambda \approx 0.68$
5.8-4	Cosmological redshift	$z = \frac{\lambda_{\mathrm{obs}} - \lambda_{\mathrm{emit}}}{\lambda_{\mathrm{emit}}}$



Hubble's law: recession velocity against distance (simulated data around the modern value).